

Grade 7
Unit 8
Lesson 3 Practice

Name _____
Date _____
Period _____

Two students evaluated the expression $\frac{(5^2 \cdot 2.8^0)^4}{5^7}$.

Patricia's Work	Kim's Work	Word Bank
$\frac{(5^2 \cdot 2.8^0)^4}{5^7}$	$\frac{(5^2 \cdot 2.8^0)^4}{5^7}$	zero
$\frac{(5^2 \cdot 1)^4}{5^7}$	$\frac{5^8 \cdot 2.8^0}{5^7}$	identity
$\frac{(5^2)^4}{5^7}$	$\frac{5^8 \cdot 1}{5^7}$	power of a power
$\frac{5^8}{5^7}$	$\frac{5^8}{5^7}$	product of powers
5^1	5^1	power of a product
5	5	quotient of powers
		power of a quotient

Complete the statements to explain each student's work by using the phrases from the word bank for questions 1-6.

- Patricia used the _____ exponent law first by rewriting 2.8^0 as 1.
- Next, she multiplied by 1 then she used the _____ exponent law by multiplying the exponents 4 and 2.
- Patricia then used the _____ exponent law by subtracting 8 and 7. Lastly, she found the value of the expression.
- Kim used the _____ exponent law first by multiplying 4 and 2 and by multiplying 4 and 0.
- Next, she used the _____ exponent law by rewriting 2.8^0 as 1.
- Kim then used the _____ exponent law by subtracting 8 and 7. Finally, she found the value of the exponential expression.

Use the laws of exponents to evaluate the expressions.

7. $\left(\frac{1.5^5}{1.5^3}\right)^2$

8. $\left[\left(\frac{4}{5}\right)^0 \cdot \left(\frac{4}{5}\right)^1\right]^2$

9. $4.7^2 \cdot (4.7^1)^3$

10. $\frac{9.9}{8.8^0} \cdot 9.9^3$